TIRUPATI WASTE-TO-ENERGY

Wet Waste & Institutional Food Waste → Compressed Biogas

Investor Presentation — Tirupati, Andhra Pradesh

Prepared by: Vi (Founder) **City Focus**: Tirupati, Andhra Pradesh, India **Technology**: Anaerobic Digestion → Compressed Biogas (CBG) **Stage**: Pre-Seed / Series A — Plant 1 Feasibility Complete **Date**: April 2026 **Confidential** — Not for distribution without founder consent

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SLIDE 1 — THE OPPORTUNITY IN ONE LINE

> ***Tirupati generates 130 tonnes of organic waste every day. 80 tonnes of it has nowhere to go. We convert it to Compressed Biogas and sell it to India's oil companies under a 15-year government contract at ₹54–72 per kg.***

SLIDE 2 — WHY TIRUPATI? THE CITY THAT NEVER STOPS GENERATING WASTE

Tirupati is not an ordinary Indian city. It is one of the world's busiest pilgrimage destinations — which means it generates organic waste at an intensity that no seasonal city can match.

City Characteristic	Fact
Resident population (2024 metro)	~775,000
Daily pilgrims to Tirumala (average)	50,000–100,000
Daily pilgrims (festivals/weekends)	Up to 300,000
Meals served free daily (TTD Annadanam)	1–2.5 lakh (Tirumala + Tirupati)
Rice consumed by TTD daily	12 tonnes
Vegetables consumed by TTD daily	7 tonnes
Hotels, dharamshalas, restaurants serving pilgrims	Hundreds
Cattle in Tirupati/Chittoor dairy belt	Hundreds of thousands
Daily milk production (Chittoor district)	18–20 lakh litres

The insight: A city with 50,000–300,000 pilgrim visitors per day produces institutional food waste that never stops. This is the most predictable, high-density organic feedstock base in Andhra Pradesh — and it is almost entirely unused for energy.

> *Source: Swachh Bharat Mission (SBM Urban); TTD official press releases; Dairy Chronicle Chittoor 2024*

SLIDE 3 — THE PROBLEM: TIRUPATI'S ORGANIC WASTE GAP

What Tirupati Generates

Waste Stream	Daily Volume (TPD)	Current Status
City wet waste (households)	115	50 TPD processed by Bio-methanation plant[1] → **65 TPD unprocessed**
City food waste	15	Partially composted, mostly landfilled
Institutional food waste (hotels, dharamshalas, pilgrim economy)	~20–30 (estimated)	No dedicated processing
Dairy cattle manure (Chittoor belt, 50 km radius)	Significant (unlocked by hub)	Open field disposal
Groundnut shells (seasonal, region)	5–10	Burned or abandoned
Organic waste with no good destination	**~80–100 TPD**	**Landfill / open burning**

[1] *Tirupati's existing Bio Green City bio-methanation plant processes 50 TPD → 1,728 m^3/day raw biogas sold locally to hotels. This is NOT CBG — it is low-grade, not SATAT-registered, and does not feed the national gas grid.*

TTD-IOCL Biogas Plant (Tirumala hilltop): 40–50 TPD → 1,500 kg/day biogas — confined to the hilltop, uses TTD food waste, feeds back to TTD kitchens only. This feedstock and project are separate from our venture.

The Consequence

- **Landfills filling**: Tirupati generates ~219 TPD total MSW; landfill capacity under pressure
- **Pilgrim economy food waste**: No bulk waste processor for hotels/restaurants — SWM Rules 2026 (effective 1 April 2026) now mandate bulk generators (>100 kg/day) to arrange processing or pay a penalty
- **Field burning**: Chittoor farmers burn groundnut shells and paddy straw seasonally
- **Lost energy**: 80–100 TPD of high-calorific organic material decomposes in landfills, releasing methane to atmosphere with zero economic return

> *Source: SBM Urban Tirupati Waste Data; SWM Rules 2026 (MoHUA)*

SLIDE 4 — OUR SOLUTION

One Plant. Four Feedstock Streams. Four Revenue Streams.

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FEEDSTOCK IN                    PLANT                  PRODUCTS OUT
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City organic waste (40 TPD)  ┐                    ┌─ CBG → OMC offtake (IOCL/HPCL)
Institutional food waste      │  ANAEROBIC         │  (15-year contract, ₹54–72/kg)
  (hotels/dharamshalas)       │  DIGESTION    ───→ │
  (20 TPD)                    │  CSTR +            ├─ Digestate → Biofertiliser
Dairy cattle manure           │  PSA Upgrading     │  (groundnut/paddy farmers,
  (Chittoor belt, 25 TPD)    │                    │   ₹5,000–8,000/tonne)
Groundnut shells (5 TPD)     ┘                    ├─ Tipping Fees (paid by waste
                                                   │  generators – TMC + hotels)
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100 TPD in → 12–14 TPD CBG out                    └─ Carbon Credits (VCS/Gold
                                                        Standard, ₹400–1,000/tCO₂e)
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Why Tirupati Is Structurally Different from Other CBG Sites

Factor	Generic India CBG Site	Tirupati
Feedstock risk	High — depends on one crop/season	Low — pilgrim economy generates waste 365 days/year
Tipping fee potential	Low — municipal waste only	High — hotels, dharamshalas under SWM 2026 mandate
Institutional anchor partner	None typically	TTD city-level partnership possible
Agricultural co-substrate	Varies	Groundnut shells (year-round, Chittoor)
CBG offtake route	Variable	HPCL Tirupati CNG station + road tanker to grid
State incentive stack	National schemes only	AP ICEP 2024 + national SATAT + MNRE CFA

SLIDE 5 — FEEDSTOCK DEEP DIVE

Stream 1: City Organic Waste (40 TPD) — Tipping Fee Model

Tirupati Municipal Corporation (TMC) already collects 130 TPD of organic/wet waste. Only 50 TPD is processed by the existing bio-methanation unit (raw biogas, not CBG). The remaining 80 TPD has no processor.

Our arrangement: TMC pays us a tipping fee to receive and process their residual organic fraction (40 TPD initially).

- Benchmark tipping fee: ₹800–1,500/tonne (Swachh Bharat Mission data; Indore ₹1,080/t)
- Revenue from TMC: ₹38–72 lakh/year at 40 TPD

SWM 2026 trigger: From 1 April 2026, bulk generators (hotels, institutions >100 kg/day) must process waste or pay Extended Producer Responsibility charges. This creates a second, parallel tipping fee customer base in Tirupati's hospitality sector.

Stream 2: Institutional Food Waste — Pilgrim Economy (20 TPD)

Tirupati hosts hundreds of hotels, dharamshalas, and restaurants serving pilgrims. Conservative estimate: 50 major establishments each generating 200–400 kg/day food waste = **10,000–20,000 kg/day = 10–20 TPD**.

TTD city-level establishments: TTD operates guest houses, rest houses, and administrative canteens in Tirupati city (not just Tirumala). These are separate from the Tirumala hilltop IOCL plant and represent a natural partnership opportunity for tipping fee or waste supply agreement.

Revenue: Institutional food waste commands ₹1,000–2,000/tonne tipping fee (Delhi benchmark ₹2,000–2,700/t; Indore ₹1,080/t). At ₹1,000/t: 20 TPD × ₹1,000 × 330 days = **₹66 lakh/year**.

Stream 3: Dairy Cattle Manure — Chittoor Belt (25 TPD)

Chittoor district (which includes Tirupati) is Andhra Pradesh's leading dairy district:

- 18–20 lakh litres daily milk collection
- ~80% farmers depend on dairy
- 2,173 Gokulam cattle sheds sanctioned under state scheme

Each litre of milk → approximately 3–4 kg wet manure. 18 lakh litres/day = **54,000–72,000 TPD manure** across the district. We need only 25 TPD — less than 0.05% of district production.

Cost: Manure is typically free or ₹0–200/tonne at source. Hub-and-spoke collection within 30 km radius. **Co-digestion value**: Cattle manure stabilises digester pH, improves biogas yield 25–50% in co-digestion with food waste (literature verified).

Stream 4: Groundnut Shells (5 TPD, Seasonal Enhancement)

Chittoor district grows significant groundnut. Groundnut shells are lignocellulosic:

- Need mild alkaline pretreatment (NaOH) before AD — standard practice
- Co-digestion with food waste at 50:50 ratio improves biomethane yield by 32% (Springer 2025 study)
- We use as seasonal yield booster, not primary feedstock
- Cost: near zero (waste material, farmer pays to dispose)

> *Sources: Swachh Bharat Mission Tirupati; SWM Rules 2026; Dairy Chronicle Chittoor 2024; Springer Nature 2025 (groundnut co-digestion); SBM Urban waste data*

SLIDE 6 — TECHNOLOGY STACK

Proven. Modular. Replicable.

We use no experimental technology. Every component has been commercially deployed in India.

Component	Choice	Why
Digester type	CSTR (Continuous Stirred Tank Reactor)	Handles mixed feedstocks (food waste + manure + shells); dominant in India; Praj, Enviropol supply locally
Operating temperature	Mesophilic (35–38°C)	Lower energy input than thermophilic; stable with Tirupati's warm climate (avg 30°C+)
Biogas upgrading	PSA (Pressure Swing Adsorption)	Phase 1 technology; no water use; widely deployed in AP/South India; upgrades raw biogas to >95% CH_4
CBG compression	CNG-standard compression train	Road tanker filling at 200–250 bar
Digestate handling	Solid-liquid separator → FCO-certified biofertiliser	Dry cake to groundnut/paddy farmers; liquid fraction for irrigation
Monitoring	IoT sensors (pH, temperature, gas yield)	Remote SCADA; VCS-compliant data archiving for carbon credits

Technology Reference: HPCL Badaun (UP) Benchmark

The HPCL-Praj plant at Badaun (commissioned January 2024) is our closest commercial comparator:

- Scale: 100 TPD feedstock (paddy straw)
- Output: 14 TPD CBG
- CAPEX: ₹133 crore gross
- Technology: RenGas™ by Praj Industries (CSTR + PSA)
- Status: Operational, supplying HPCL under SATAT

We adapt this proven configuration for Tirupati's multi-feedstock mix (food waste + manure + shells vs. straw). Multi-feedstock actually **reduces technology risk** — no single feedstock failure shuts the plant.

Why Not Other Technologies?

Alternative	Why We Reject for Phase 1
Biomass gasification (thermal)	High moisture feedstock (food waste) incompatible; requires dry feedstock
Direct combustion / incineration	High CAPEX; regulatory complexity; public opposition in pilgrim city
Landfill gas capture	Tirupati landfill too small; gas quality variable
Composting only	No energy revenue; low tipping fee; not CBG

SLIDE 7 — MARKET CONTEXT: WHY NOW IN INDIA, WHY NOW IN AP

National Tailwinds (Already Documented)

Driver	Status	Impact on Tirupati
SATAT Scheme	1,094 LoIs issued; only 108 commissioned (10%)	Tirupati plant would be among first operational in AP
Compressed Biogas Obligation (CBO)	Mandatory 1% blend FY2026, rising to 5% FY2029	OMC (HPCL/IOCL) must buy CBG — guaranteed demand
SWM Rules 2026	Bulk generators >100 kg/day must process or pay	Creates tipping-fee customers in Tirupati's hotel/pilgrimage economy immediately
CBG price revision (May 2025)	85% of CNG retail price (up from 80%)	₹62–72/kg realistic at Tirupati CNG stations

Andhra Pradesh — State-Level Advantages

AP Incentive	Detail	Value to Project
AP Integrated Clean Energy Policy 2024	Approved by state cabinet	Legal framework for clean energy investment
Revenue land lease	₹15,000/acre/year (5% biennial escalation)	Low land cost vs. outright purchase
SGST reimbursement	5 years from commissioning	~₹1–2 crore/year tax benefit

Electricity duty waiver	Full waiver for green energy projects	~₹30–50 lakh/year savings on plant power
Capital subsidy	Biofuel project category	Additional grant on top of national MNRE CFA
Reliance's 500-plant programme	Validates AP as CBG state	Creates supply chain, trained contractors, policy priority

The Reliance Factor: Risk or Opportunity?

Reliance has committed ₹65,000 crore for 500 CBG plants across AP. Their **first plant at Kanigiri (Prakasam district)** uses Napier grass — an energy crop grown on 475 acres of barren land.

This is a fundamentally different model from ours:

Dimension	Reliance (Kanigiri)	Our Venture (Tirupati)
Feedstock	Napier grass (energy crop, 475 acres)	Urban/institutional waste + dairy manure
Land requirement	475 acres minimum per plant	5–10 acres
Feedstock cost	Energy crop cultivation cost	₹0 tipping fee (waste generators pay us)
Scalability constraint	Land + water	Waste volume (grows with city/pilgrims)
Revenue model	CBG only	CBG + tipping fee + digestate + carbon
Competition risk	Direct on feedstock? No — different source	None — they need large barren land, we need none

Conclusion: Reliance validates the AP CBG market and improves state-level policy priority. They do not compete for urban organic waste or Tirupati's pilgrim economy feedstock.

> *Sources: AP ICEP 2024 (ChiniMandi); Reliance AP CBG groundbreaking press release April 2025; SATAT IOCL overview; PIB CBG data*

SLIDE 8 — REVENUE MODEL

Four Revenue Streams from One Plant

Plant Specification: 100 TPD feedstock → 12–14 TPD CBG → 330 operating days/year

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Revenue Stream 1: CBG Sales to OMC (Primary Revenue)

Parameter	Conservative	Base Case	Optimistic
Daily CBG output	11 TPD	13 TPD	14 TPD
Utilisation rate (Year 3)	85%	92%	98%
Effective selling price	₹54/kg (SATAT floor)	₹62/kg (market)	₹70/kg (revised CBO rate)

Annual CBG revenue	₹16.6 crore	₹24.3 crore	₹31.4 crore

SATAT 15-year OMC offtake agreement with HPCL/IOCL is the anchor revenue. Floor price ₹46/kg guaranteed; market rate ₹54–72/kg.

Revenue Stream 2: Tipping Fees (Waste Generators Pay Us)

Customer Type	Volume (TPD)	Rate (₹/tonne)	Annual Revenue
Tirupati Municipal Corporation	40	₹900	₹1.19 crore
Hotels/Dharamshalas/Restaurants (SWM 2026 mandate)	20	₹1,200	₹79 lakh
TTD city establishments (partnership)	10	₹500 (discounted for anchor partner)	₹17 lakh
Total Tipping Fee Revenue	**70 TPD**	—	**₹2.15 crore/year**

SWM Rules 2026 effective 1 April 2026: bulk generators >100 kg/day must arrange processing or pay EBWGR charges. This creates immediate demand for our tipping fee offer.

Revenue Stream 3: Digestate Biofertiliser

Parameter	Value
Digestate output	~35 TPD solid + ~80 TPD liquid per 10 TPD CBG produced
Solid digestate (FCO-certified)	25 TPD dry biofertiliser
Selling price	₹5,000–8,000/tonne (to groundnut, paddy, sugarcane farmers in Chittoor)
Annual digestate revenue (base)	25 TPD × ₹6,000 × 330 days = **₹4.95 crore**

Chittoor district grows groundnut, paddy, sugarcane — all high-value crops where organic fertiliser commands premium. Direct farm supply within 50 km radius eliminates distribution cost.

Revenue Stream 4: Carbon Credits

Parameter	Value
Emissions avoided per year	~35,000–45,000 tCO_2e (methane capture + fossil fuel displacement)
Carbon credit price (VCM, 2024)	₹200–400/tCO_2e
Carbon credit price (compliance from 2026)	₹800–1,000/tCO_2e
Annual carbon revenue (VCM base case)	40,000 × ₹300 = **₹1.2 crore**

Annual carbon revenue (compliance, Year 3+)	40,000 × ₹800 = **₹3.2 crore**

Year 3 Revenue Summary

Revenue Stream	Base Case	% of Total
CBG Sales (SATAT/OMC)	₹24.3 crore	71%
Tipping Fees	₹2.15 crore	6%
Digestate Biofertiliser	₹4.95 crore	15%
Carbon Credits	₹1.2 crore	4%
Total Revenue (Year 3)	**₹32.6 crore**	100%

> *Sources: SATAT IOCL pricing; GOBARdhan CBG procurement price 2024; SBM Urban tipping fee benchmarks; Mongabay India digestate market 2023; Gold Standard/VCS carbon market data*

SLIDE 9 — UNIT ECONOMICS & FINANCIAL MODEL

Capital Expenditure (100 TPD Plant, Tirupati)

Component	Cost (₹ crore)
Land preparation (5–8 acres; AP leased at ₹15,000/acre/year)	0.5
Civil construction (digester hall, admin, utilities)	18–22
CSTR digesters (2 × 2,000 m^3)	20–25
PSA biogas upgrading unit	12–15
CBG compression + storage + dispensing	8–10
Feedstock pre-processing (shredder, screener, mixer)	5–7
Digestate solid-liquid separation + storage	4–5
Electrical + instrumentation + SCADA	6–8
Contingency (10%)	8–10
Gross CAPEX	**₹95–107 crore**
Less: MNRE Central Financial Assistance (grant)	–₹10 crore
Less: AP State Capital Subsidy (biofuel category)	–₹3–5 crore
Net CAPEX (Investor + Debt Burden)	**₹80–97 crore**

Operating Expenditure (Year 3 Steady State)

Item	Annual Cost (₹ crore)
Feedstock (net of tipping fee income)	**₹0 — tipping fee model reverses this to REVENUE**
Operations & maintenance (plant)	3.0–4.0
Manpower (30–35 staff)	2.5–3.5
Utilities (power, water — AP electricity duty waiver year 1–5)	1.2–1.8
Feedstock logistics (hub-and-spoke, 30 km radius)	0.8–1.2
Carbon credit verification (VCS)	0.3–0.5
Insurance, land lease, admin	0.5–0.8
Total OPEX	**₹8.3–11.8 crore/year**

Key structural advantage: Tipping fees convert feedstock from cost-centre to profit-centre. This eliminates the primary failure mode of India CBG plants (feedstock price spikes).

P&L Summary

	Year 1	Year 2	Year 3
Total Revenue	₹14.5 crore	₹24.0 crore	₹32.6 crore
OPEX	₹9.5 crore	₹10.5 crore	₹11.0 crore
EBITDA	**₹5.0 crore**	**₹13.5 crore**	**₹21.6 crore**
EBITDA Margin	**34%**	**56%**	**66%**
Depreciation	4.5 crore	4.5 crore	4.5 crore
Interest (IREDA loan, ~10%)	6.0 crore	5.8 crore	5.5 crore
PAT	**–₹5.5 crore**	**₹3.2 crore**	**₹11.6 crore**

Year 1 loss is normal for greenfield CBG (utilisation ramp-up). Break-even at ~60% utilisation (Month 14–16).

Funding Structure

Source	Amount	Terms
MNRE CFA Grant	₹10 crore	Non-repayable; disbursed 30/40/30 at milestones
AP State Capital Subsidy	₹3–5 crore	Non-repayable
IREDA Priority Sector Loan	₹55–60 crore	~10% p.a.; 15-year tenor; CBG-specific scheme
Equity (Series A — This Round)	**₹25 crore**	Founder + Series A investors
Total Project Cost (Gross CAPEX)	**~₹97 crore**	

Investor Returns

Metric	Value
Project IRR (unlevered)	18–24%
Equity IRR (levered, ₹25 crore equity)	28–38%
Payback period (equity)	3.5–4.5 years
DSCR (Year 3 at IREDA terms)	2.2–2.6×
Year 5 EBITDA (Tirupati Plant alone)	~₹22–24 crore
Exit EV at 8× EBITDA (single plant)	**₹176–192 crore**
3-Plant Portfolio EV (Year 5–6)	**₹500–600 crore**

Stress Test: What Happens If Things Go Wrong?

Scenario	Revenue Impact	EBITDA Impact	Project Still Viable?
CBG price drops to SATAT floor (₹46/kg)	–₹5.5 crore	–₹5.5 crore	Yes — IRR falls to 14%
Utilisation only 70% in Year 3	–₹6.8 crore	–₹6.8 crore	Yes — DSCR 1.4×
No carbon credit (Year 1–2)	–₹1.2 crore	–₹1.2 crore	Yes — minor impact
Tipping fee reduced to ₹500/t	–₹1.1 crore	–₹1.1 crore	Yes — digestate compensates
All four simultaneously	**–₹14.6 crore**	**–₹14.6 crore**	**No — project DSCR <1.0×**

Triple+ simultaneous downside has no historical precedent in India's CBG sector. Single and dual downside scenarios remain investable.

> *Sources: HPCL Badaun CAPEX benchmark (Praj Industries / BioenNews 2024); IREDA CBG financing scheme; MNRE CFA guidelines; SATAT pricing; Global Methane Initiative CBG report January 2025*

SLIDE 10 — COMPETITIVE POSITION IN ANDHRA PRADESH

Who Is Operating CBG in AP Today?

Player	Status in AP	Feedstock	Geography	Threat to Us?
Reliance Industries	500 plants planned; Plant 1 (Kanigiri) groundbreaking April 2025	Napier grass (energy crop)	Prakasam, rural districts	No — different feedstock, no urban waste

IOCL-TTD	Tirumala plant under construction (₹11.85 crore)	Temple food waste (hilltop only)	Tirumala hilltop	No — confined to TTD hilltop; we partner with their city-level waste
EverEnviro	Strong in eastern India; limited AP presence	MSW + agricultural	Not focused on Tirupati	Low — geography gap
GPS Renewables	Mission 1111 (all-India)	MSW + food waste	Urban cities	Medium — but no Tirupati plant
HPCL	24-plant national programme	Agricultural straw	Rural/peri-urban	Low — straw focus, not urban organic
SATAT AP commissioned plants	**1 (as of early 2026)**	—	—	**Near-zero competition in Tirupati**

Our Competitive Moat in Tirupati

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LAYER 1 — FEEDSTOCK EXCLUSIVITY
 First SATAT-registered CBG plant in Tirupati = first mover for city organic waste
tipping fee contracts
 TMC exclusive processing agreement + SWM 2026 mandate locks in city supply

LAYER 2 — INSTITUTIONAL ANCHOR
 TTD partnership (city-level waste from guesthouses/establishments) = anchor supply
agreement
 300,000-pilgrim-day demand ceiling creates permanent institutional food waste floor

LAYER 3 — MULTI-STREAM REVENUE
 Tipping fee + CBG + digestate + carbon = 4 revenue streams vs. CBG-only competitors
 Any single stream collapse does not threaten the business

LAYER 4 — CHITTOOR DAIRY MANURE HUB
 First aggregator in Chittoor cattle belt = preferred manure buyer for 30 km radius
dairy farmers
 Manure = free or near-free feedstock that competitors using energy crops can never
match on cost

LAYER 5 — AP POLICY STACK
 AP ICEP 2024 incentives (SGST, electricity, land) + national SATAT + MNRE grant
 First entrant captures the easiest subsidy window; late entrants face degression

LAYER 6 — DIGESTATE MARKET LOCK-IN
 Groundnut and paddy farmers in Chittoor signed to multi-year digestate supply
agreements
 = sticky customer base, recurring revenue, natural barrier against digestate disposal
risk
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SLIDE 11 — IMPLEMENTATION PLAN

Timeline: Month 0 to Month 24 (First CBG to OMC)

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MONTH    MILESTONE
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M0       Incorporation + land selection (AP revenue land, lease agreement)
         SATAT LoI application to HPCL/IOCL
         Begin feedstock supply negotiations (TMC + hotels + dairy co-operatives)

M1–M3    Regulatory clearances:
         → Consent to Establish (CTE) from APPCB
         → Environmental Clearance (EC) if >10,000 TPY
         → PESO approval (gas handling)
         → Local municipality NOC
         MNRE CFA application submitted

M3–M6    DPR (Detailed Project Report) finalised
         IREDA loan application submitted with DPR
         EPC vendor shortlisted (Praj RenGas™ / Enviropol / GPE)
         IREDA loan sanction (target: Month 5)
         MNRE CFA tranche 1 (30%) received

M6–M12   Civil construction (digester foundations, civil works)
         Equipment procurement + delivery (CSTR, PSA, compression)
         Feedstock supply agreements signed:
         → TMC tipping fee agreement (long-term)
         → Top 30 hotels/dharamshalas (SWM 2026 mandate leverage)
         → 5 dairy co-operatives in Chittoor (30 km radius)
         Hub-and-spoke logistics fleet commissioned (5–8 vehicles)

M12–M18  Equipment installation + commissioning
         Biogas plant trial runs (raw biogas production begins M15)
         PSA upgrading unit commissioned (CBG quality testing)
         MNRE CFA tranche 2 (40%) received
         SATAT LoI converted to formal offtake agreement
         Staff training completed (local hire + Praj commissioning team)

M18      ─── FIRST CBG DELIVERED TO OMC ───
         Road tanker dispatch to HPCL CNG station (Tirupati)
         Revenue recognition begins

M18–M24  Ramp-up to 85% utilisation
         Carbon credit baseline documentation (VCS methodology)
         Digestate FCO certification obtained
         Digestate farmer agreements in place
         MNRE CFA tranche 3 (30%) received

M24      ─── FULL UTILISATION — INVESTOR RETURN CLOCK STARTS ───
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Key Regulatory Approvals (Tirupati-Specific)

Approval	Issuing Authority	Timeline	Status at Year 1
Consent to Establish (CTE)	APPCB (AP Pollution Control Board)	60–90 days	Critical path item
SATAT LoI	HPCL or IOCL (Tirupati)	30–60 days after application	Determines offtake buyer
Environmental Clearance	MoEFCC / SEIAA	90–120 days	Required if scale >threshold
PESO Certificate	Petroleum & Explosives Safety Organisation	60–90 days	Gas storage and compression
MNRE CFA	MNRE (New Delhi)	90 days (application to approval)	Unlocks ₹10 crore grant
AP Revenue Land Lease	District Collector, Tirupati	45–60 days	Parallel track with CTE
Local NOC	Tirupati Municipal Corporation	30 days	Early engagement advised

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SLIDE 12 — TEAM & EXECUTION MODEL

Key Roles (30–35 Staff, Plant 1)

Role	Hire Profile	Criticality
CEO / Founder (Vi)	Strategy, investor relations, government liaison	Core
Plant Manager	Chemical/environmental engineer, 5+ years biogas	Critical
Feedstock & Logistics Manager	Supply chain background, local networks in Tirupati/Chittoor	**Most Critical Hire**
Process Control Engineer	Biogas operations, SCADA	Critical
Digestate & Agri Sales Manager	Agri-business, Chittoor farmer networks	High
Finance & Regulatory Manager	AP regulatory experience, IREDA/MNRE familiarity	High
Operations Staff (25 workers)	Local hire, trained by EPC vendor	Standard

Execution Philosophy

1. **Tipping fee first** — sign TMC and hotel contracts before breaking ground; these de-risk feedstock
2. **Hire the Feedstock Manager in Month 1** — this role determines plant utilisation more than any technology decision

3. **EPC with commissioning guarantee** — Praj RenGas™ or equivalent; performance bond tied to CBG quality specs
4. **Digestate before commissioning** — farmer agreements signed before first digestate flows; no disposal crisis
5. **APPCB early engagement** — CTE delays are the most common India CBG plant delay; start regulatory conversations in Month 0

Partnership Strategy

Partner	Role	Status
TTD (city level)	Anchor waste supply + reputational endorsement	To be negotiated in Month 0–3
Tirupati Municipal Corporation	Tipping fee + waste supply agreement	Leverage SWM 2026 mandate
HPCL or IOCL Tirupati	SATAT offtake (15 years)	Apply SATAT LoI Month 0
Praj Industries / Enviropol	EPC + technology license	Shortlist by Month 3
IREDA	Debt (₹55–60 crore)	Apply Month 3 with DPR
Chittoor Dairy Co-operatives	Manure supply (free/low cost)	Negotiate Month 3–6
Andhra Pradesh New & Renewable Energy Development Corporation (NREDAP)	State subsidy facilitation	Early registration

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SLIDE 13 — POLICY ALIGNMENT & PUBLIC VALUE

National + AP Policy Stack

Scheme	Applicable?	Benefit
SATAT (IOCL/HPCL)	Yes	15-year offtake, ₹46–72/kg floor
MNRE Central Financial Assistance	Yes	₹10 crore non-repayable grant
Compressed Biogas Obligation (CBO)	Yes	Mandates OMC demand, improves pricing
SWM Rules 2026	Yes	Creates tipping-fee customer base
IREDA Priority Sector Lending	Yes	₹55–60 crore at ~10% p.a., 15-year tenor
GOBARdhan scheme	Yes	Biofertiliser FCO certification pathway
AP ICEP 2024	Yes	SGST reimbursement (5 yr), land lease, electricity waiver

AP Revenue Land (₹15,000/acre/yr)	Yes	Near-zero land cost vs. acquisition
PM KUSUM (for farmer linkage)	Partial	Supports farmer adoption of digestate

Seven Public Value Outcomes Per Plant

Outcome	Impact in Tirupati	Audience
Organic waste diverted from landfill	80–100 TPD → zero landfill	Government, Citizens
Pilgrimage city cleanliness	Institutional food waste processor for SWM 2026 compliance	TMC, TTD, Government
Clean fuel for transport	12–14 TPD CBG replaces diesel/CNG	OMC, Citizens
Chittoor farmers get biofertiliser	Reduces urea dependency; FCO-certified	Farmers, Government
GHG emissions avoided	35,000–45,000 tCO_2e/year per plant	Climate, Government
Local employment	30–35 direct; 50–100 indirect (logistics, agri)	Citizens, Government
Straw/shell burning stopped	Groundnut shells, paddy residue → feedstock not smoke	Citizens, Health

The Tirupati Narrative for Government

> *"Tirupati receives 3–10 crore devotees per year. Their visit generates waste that currently goes to landfills or clogs drains. This plant converts that waste to clean fuel that powers the same CNG vehicles bringing pilgrims to the city. Tirupati becomes a circular pilgrimage economy — the first in India."*

This narrative works for:

- **State Government (AP)**: Investment, employment, environmental compliance, clean energy targets
- **Tirupati Municipal Corporation**: SWM 2026 compliance, Swachh Survekshan ranking maintenance (currently 8th cleanest city nationally)
- **TTD**: Waste management for the city's pilgrim economy; environmental stewardship
- **PNGRB / MoPNG**: AP CBG market development, SATAT execution rate improvement

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SLIDE 14 — RISKS & MITIGATION

Risk Register (Tirupati-Specific)

#	Risk	Probability	Impact	Mitigation
1	**Feedstock supply below 80%** (city waste diversions, hotel low season)	Medium	High	4-stream mix; manure + shells buffer seasonal dips; pilgrimage never stops

2	**TMC tipping fee agreement delayed**	Medium	Medium	Begin negotiations Month 0; SWM 2026 mandate is leverage
3	**TTD partnership not formalised**	Medium	Low	TTD city waste is one of 4 streams — not a single point of failure
4	**APPCB CTE delay**	Medium	High	Engage APPCB in Month 0; use AP ICEP fast-track for green energy projects
5	**CBG quality rejection by OMC**	Low	High	PSA technology delivers >95% CH_4 consistently; Praj performance bond
6	**Digestate disposal crisis** (no farmer buyers)	Medium	Medium	Sign farmer agreements BEFORE plant commissioning; FCO certification pre-commission
7	**IREDA loan delay**	Low	High	Parallel application to SIDBI CBG; bank finance fallback
8	**Reliance enters Tirupati**	Low	Medium	Reliance uses energy crops (475+ acres needed); Tirupati has no barren land at that scale
9	**CGD pipeline not operational**	Medium	Low	Road tanker offtake to HPCL/IOCL CNG stations; no pipeline dependency for Phase 1
10	**Groundnut shell pretreatment cost**	Low	Low	Only 5 TPD — minor cost; simple NaOH soak method

What Breaks the Thesis (Triple Downside)

The investment thesis breaks only if ALL of the following occur simultaneously:

1. CBG price falls to ₹40/kg (below SATAT floor — requires government policy reversal)
2. Utilisation stays below 55% for 24+ months (requires all 4 feedstock streams to fail simultaneously)
3. Digestate has zero market (requires AP groundnut/paddy farming to collapse)

No historical precedent for this combination in India's CBG sector.

SLIDE 15 — THE ASK

Series A: ₹25 Crore Equity

Use of Funds	Amount
CAPEX equity contribution (supplement IREDA debt)	₹18 crore
Working capital (Year 1–2 operations pre-full revenue)	₹4 crore
Regulatory + legal + EPC procurement costs	₹2 crore

Team hiring (Plant Manager, Feedstock Manager, Finance)	₹1 crore
Total Series A Ask	**₹25 crore**

What ₹25 Crore Buys

- Plant 1 in Tirupati fully funded to commissioning (Month 18)
- First CBG to HPCL/IOCL under SATAT (Month 18)
- Full utilisation and positive EBITDA (Month 24)
- Platform for Plants 2 and 3 (Vijayawada / Visakhapatnam, Year 3)

Investor Return Scenarios (₹25 Crore Equity In)

Exit Scenario	Year	Exit Event	Equity Value	Return Multiple
Single-plant sale	Year 4	Plant 1 EBITDA ₹20 crore × 8×	₹160 crore	6.4×
3-plant portfolio	Year 6	EBITDA ₹60 crore × 8× = ₹480 crore EV; equity ~40%	₹192 crore	7.7×
OMC acquisition	Year 5–6	HPCL/IOCL acquires operating CBG platform in AP	Strategic premium 10–12×	10–12×
Infrastructure fund	Year 6–7	Yield-seeking capital; 7–9× EBITDA	~₹160–240 crore equity	6.4–9.6×

SLIDE 16 — 90-DAY ACTION PLAN

First 90 Days After Funding

Week	Action	Owner
Week 1–2	Incorporate operating entity (AP); open IREDA-eligible entity bank account	Founder + Legal
Week 1–2	Apply for SATAT LoI to HPCL Tirupati division	Founder
Week 2–4	Identify land (revenue land: 5–8 acres, 10–15 km from Tirupati city)	Founder + AP revenue dept
Week 3–6	Hire Plant Manager + Feedstock & Logistics Manager	Founder
Week 4–8	Begin feedstock negotiation: TMC (tipping fee), top 30 hotels, Chittoor dairy co-ops	Feedstock Manager
Week 4–8	Engage APPCB for CTE pre-consultation	Plant Manager + Legal
Week 6–10	Commission DPR (Detailed Project Report) from EPC vendor	Plant Manager
Week 8–12	Submit MNRE CFA application	Finance Manager

Week 8–12	Submit IREDA loan application (with DPR)	Finance Manager
Week 10–12	Sign first feedstock MoU (TMC or dairy co-op) — proof of concept for investors	Feedstock Manager
Week 12	**Investor progress report: LoI in hand + land selected + DPR complete**	Founder

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SLIDE 17 — THE TIRUPATI THESIS IN ONE PAGE

Hypothesis

Tirupati generates 130 TPD of organic waste from a pilgrim economy that never stops. Converting this waste to Compressed Biogas creates a permanent resource from a permanent liability — and the government will pay for the output under a 15-year contract.

Business Thesis

- **Feedstock**: Urban organic waste + institutional food waste (hotels/TTD city) + Chittoor dairy manure → 100 TPD consistent, year-round supply
- **Technology**: CSTR anaerobic digestion + PSA upgrading → CBG (proven at Badaun, Praj benchmark)
- **Revenue**: 4 streams (CBG + tipping fee + digestate + carbon) → ₹32 crore/year by Year 3
- **Policy**: SATAT 15-year OMC offtake + MNRE grant + AP ICEP 2024 incentives
- **Moat**: First SATAT plant in Tirupati + TTD city partnership + Chittoor dairy manure exclusivity + digestate farmer lock-in

Why Now

1. **SWM Rules 2026 (1 April 2026)**: Hotels, dharamshalas, large kitchens must process or pay — our tipping fee offer is immediately relevant
2. **CBO mandate**: OMCs must buy CBG at 1% → 5% blend; Tirupati HPCL CNG demand is growing with pilgrimage traffic
3. **AP ICEP 2024**: State government is actively courting CBG investment; first entrants get maximum incentives
4. **Only 1 SATAT plant in AP**: Gap is wide open; Tirupati has the richest urban organic feedstock base in the state

The Numbers That Matter

Investment required	₹25 crore equity
Revenue Year 3	₹32.6 crore
EBITDA Year 3	₹21.6 crore (66%)
Equity IRR	28–38%
Exit EV (Year 5–6, 3 plants)	₹500–600 crore

The Ask

> ***₹25 crore Series A equity to commission India's first multi-feedstream CBG plant in Tirupati — converting the pilgrimage economy's waste into clean fuel and proving the model for replication across Andhra Pradesh.***

APPENDIX — DATA SOURCES

Claim	Source	Date	Confidence
Tirupati MSW: 115 TPD wet + 15 TPD food waste	Swachh Bharat Mission Urban (SBM Urban)	2023	High
Existing bio-methanation: 50 TPD → 1,728 m^3/day	SBM Urban Tirupati page	2023	High
TTD-IOCL Tirumala plant: ₹11.85 crore, 40-50 TPD, 1,500 kg/day	Deccan Chronicle; iamrenew; Energetica India	Nov 2024	High
TTD Annadanam: 1–2.5 lakh meals/day; 12T rice + 7T vegetables	TTD official press release; news.tirumala.org	2024	High
Tirupati pilgrims: 50,000–100,000/day; 300,000 peak	TTD official; Wikipedia	2024	High
Chittoor dairy: 18–20 lakh litres/day milk	Dairy Chronicle 2024; Hans India	2024	High
Tirupati Swachh Survekshan rank: 8th cleanest (>1 lakh)	NE India Broadcast; The Print	Feb 2024	High
SATAT: 1,094 LoIs, 108 commissioned (10%)	Global Methane Initiative Jan 2025	2025	High
AP: 1 SATAT plant commissioned	Global Methane Initiative; SATAT data	2025	High
Reliance Kanigiri: ₹139 crore, 100 TPD, Napier grass, 475 acres	Reliance press release; Republic World	Apr 2025	High
Reliance AP: 500 plants, ₹65,000 crore, AP ICEP approved	Business Standard; iamrenew	2025	High
AP ICEP 2024: land ₹15,000/acre/yr, SGST 5yr, electricity duty waiver	ChiniMandi; iamrenew	2024	High
CBO mandate: 1% FY2026 → 5% FY2029	PIB; PPAC; MoPNG	2024	High
SWM Rules 2026: bulk generators >100 kg/day mandatory	Vision IAS; MoHUA notification	Apr 2026	High
SATAT CBG price revised May 2025: 85% of CNG retail	iamrenew; PPAC	2025	High
MNRE CFA: up to ₹10 crore per plant	MNRE Biogas programme	2024	High

IREDA: ~10% p.a., 15-year tenor, CBG-specific	IREDA CBG scheme document	2024	High
Groundnut co-digestion: 32% biomethane improvement at 50:50	Springer Nature (Waste & Biomass Valorization)	2025	High
Tipping fee benchmark: Indore ₹1,080/t; Delhi ₹2,000-2,700/t	SBM Urban; Mongabay	2023	Medium
HPCL Badaun: ₹133 crore / 100 TPD / 14 TPD CBG	Praj Industries; BioEnergy News	2024	High
Carbon credit VCM rate: ₹200–400/tCO_2e	Gold Standard; VCS market data	2024	Medium
GAIL Kondapalli–Tirupati pipeline: 450 km, 4 MMSCMD, EOI 2020	ChemAnalyst	2020	Low (status unclear)

Presentation prepared April 2026. All figures in Indian Rupees unless stated. USD conversion at ₹83/USD. Confidential — prepared for investor discussion only. Not for public distribution. Contact: Vi (Founder) | Research base: `/home/vi/Sustinable_Energy/research/`